

StudentName[]

StudentMark[]

Asjad
Yahya
Mustafa
Maaz
Abdullah
Ahsan
Mudassir

98	97	91	286	95
48	59	69	176	58
69	67	54	190	63
29	47	39	115	38
69	64	73	206	68
67	69	94	230	76
78	81	39	198	66

ClassSize ← 7

SubjectNo ← 3

Average mark	Grade awarded
greater than or equal to 70	distinction
greater than or equal to 55 and less than 70	merit
greater than or equal to 40 and less than 55	pass
less than 40	fail

Scenario Based

#Arrays declaration

```
studentname=["Asjad","Yahya","Mustafa","Maaz","Abdullah","Ahsan","Mudassir"]
```

```
studentmarks=[[89,97,79],[48,59,69],[69,67,54],[29,47,39],[69,64,73],[67,69,94],[78,81,39]]
```

#Variable declaration

```
ClassSize=7
```

```
SubjectNo=3
```

#Counters

```
num_distinctions=0
```

```
num_merits=0
```

```
num_passes=0
```

```
num_fails=0
```

#Constants declaration

```
Distinction_marks=70
```

```
Merit_marks=55
```

```
Pass_Marks=40
```

#loop to iterate 7 times

for i in range(ClassSize):

#total all three marks

total_marks=sum(studentmarks[i])

#average of marks

average_mark=round(total_marks/SubjectNo)

#checking condition for distinction merir pass or fail

if average_mark >= Distinction_marks:

grade="Distinction"

num_distinctions+=1

elif average_mark >= Merit_marks:

grade="Merit"

num_merits+=1

elif average_mark >= Pass_Marks :

grade="Pass"

num_fails+=1

else:

grade="Fail"

num_fails+=1

#output values

print("Name",studentname[i],"\\tMarks",total_marks,"\\tAverage_Marks:",average_mark,"\\tGrade",grade)

print("Distinctions:",num_distinctions)

print("Merits:",num_merits)

print("Passes:",num_passes)

print("Fails:",num_fails)

Output

Name Asjad Marks 265 Average_Marks: 88 Grade Distinction

Distinctions: 1

Merits: 0

Passes: 0

Fails: 0

Name Yahya Marks 176 Average_Marks: 59 Grade Merit

Distinctions: 1

Merits: 1

Passes: 0

Fails: 0

Name Mustafa Marks 190 Average_Marks: 63 Grade Merit

Distinctions: 1

Merits: 2

Passes: 0

Fails: 0

Name Maaz Marks 115 Average_Marks: 38 Grade Fail

Distinctions: 1

Merits: 2

Passes: 0

Fails: 1

Name Abdullah Marks 206 Average_Marks: 69 Grade Merit

Distinctions: 1

Merits: 3

Passes: 0

Fails: 1

Name Ahsan Marks 230 Average_Marks: 77 Grade Distinction

Distinctions: 2

Merits: 3

Passes: 0

Fails: 1

Name Mudassir Marks 198 Average_Marks: 66 Grade Merit

Distinctions: 2

Merits: 4

Passes: 0

Fails: 1